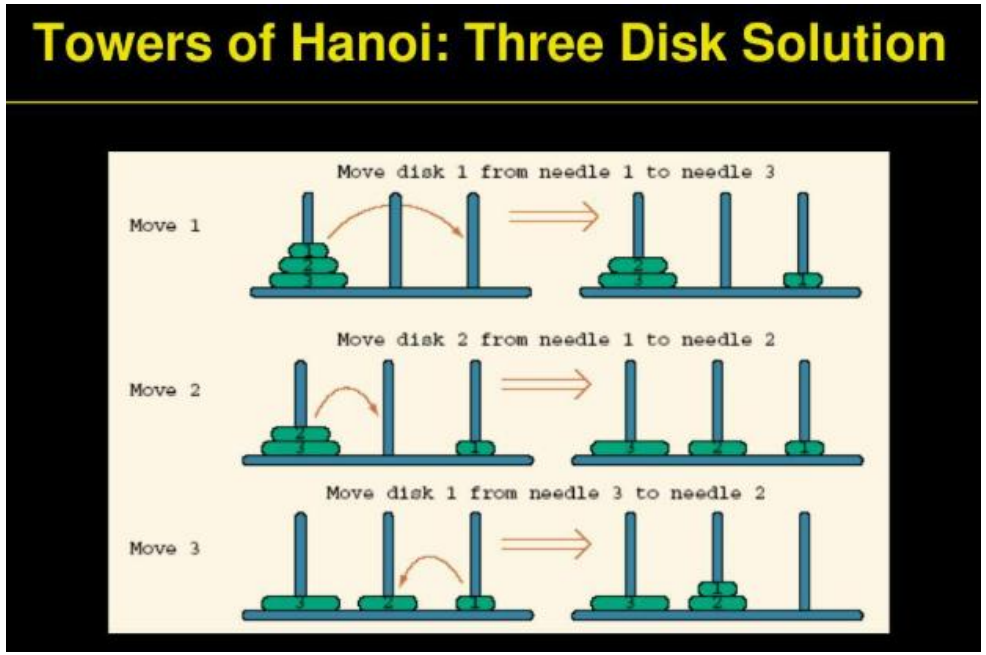
AMCS

Practice 1
Recurrent Problems 1.
The Tower of Hanoi.
Lines in the Plane.

Sandugash Kereyevna

Tower of Hanoi



Problem 1

Given

$$T_0 = 0,$$

$$T_n = 2T_{n-1} + 1,$$

find

$$T_1, T_2, T_3, T_4, T_5, T_6.$$

Find the closed form of T_n.

Tower of Hanoi

Problem 1

Given

$$T_0 = 0,$$

$$T_n = 2T_{n-1} + 1,$$

find

$$T_1, T_2, T_3, T_4, T_5, T_6.$$

Find the closed form of T_n.

Problem 2

Direct moves between pegs A and C are forbidden. Every move must be made between A and B, or between B and C.

Let H_n be the minimum number of moves required to transfer n disks from A to C.

Find

$$H_1,\ H_2,\ H_3,$$

derive a recurrence relation for H_n, and find its closed form.

Problem 3

Consider the recurrence

$$T_n = 2T_{n-2} + 2.$$

Determine whether it can describe the minimum number of moves in the Tower of Hanoi problem. Find the smallest n for which it fails.

Problem 6

Let F_n be the minimum number of moves required to transfer n disks using four pegs.

For $1 \leq k < n$, use

$$F_n \leq 2F_k + T_{n-k}.$$

Find the best upper bounds for

$$F_4, \qquad F_5.$$

1. Basic Recurrence Relations

Problem 1

Given

$$a_0 = 4, \qquad a_1 = 2,$$

$$a_n = a_{n-2} - 3a_{n-1} + 7, \qquad n \geq 2,$$

find

$$a_2,\ a_3,\ a_4,\ a_5.$$

Problem 2

Given

$$A_0 = 3,$$

$$A_n = 5A_{n-1} + 2, \qquad n \geq 1,$$

prove by induction that

$$A_n = \frac{7}{2}5^n - \frac{1}{2}.$$

Problem 3

Find the closed form of

$$B_0 = 7,$$

$$B_n = 3B_{n-1} - 6.$$

Problem 4

Find the closed form of

$$C_0 = 4,$$

$$C_n = 7C_{n-1} + 1.$$

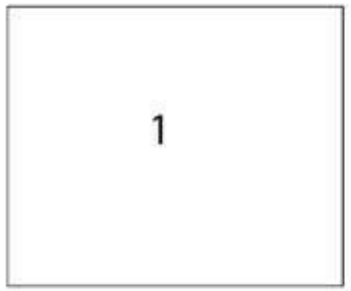
Problem 5

Find the closed form of

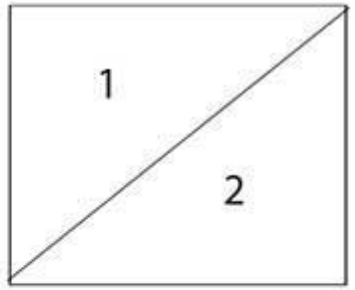
$$D_0 = 3,$$

$$D_n = 6D_{n-1} - 11.$$

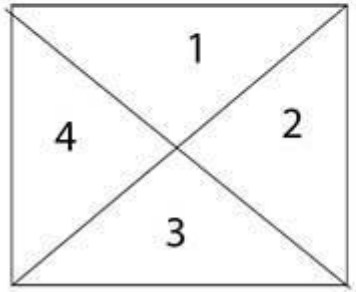
Lines in the Plane



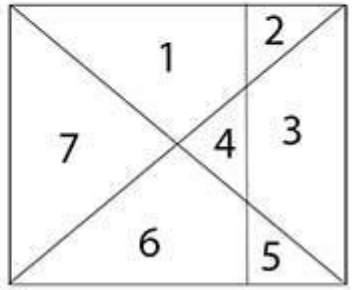
Lines[0]=1 region



Lines[1] = 2 regions



Lines[2] = 4 regions



Lines[3] = 7 regions

Problem 1

Given

$$L_0 = 1,$$

$$L_n = L_{n-1} + n,$$

find

$$L_1, L_2, L_3, L_4, L_5, L_6.$$

Find the closed form of L_n.

Lines in the Plane

Problem 1

Given

$$L_0 = 1,$$

$$L_n = L_{n-1} + n,$$

find

$$L_1, L_2, L_3, L_4, L_5, L_6.$$

Find the closed form of L_n.

Problem 2

Suppose $n-1$ lines are already drawn in general position.

The n-th line intersects all previous lines at distinct points.

Find:

$$\text{number of intersection points},$$

$$\text{number of segments of the new line},$$

$$\text{number of new regions}.$$

Hence obtain a recurrence relation for L_n.

Problem 3

Let P_n be the number of regions formed by n parallel lines.

Find

$$P_0, P_1, P_2, P_3, P_4,$$

derive a recurrence relation for P_n, and find its closed form.

Problem 4

Let C_n be the number of regions formed by n lines passing through one common point.

Find

$$C_1, C_2, C_3, C_4,$$

derive a recurrence relation for C_n, and find its closed form.

Problem 5

Suppose $n-1$ lines are in general position.

The n-th line intersects every previous line and passes through exactly one existing intersection point of two old lines.

Find the number of new regions.

Generalize the result when the new line passes through r existing intersection points.

Lines with a Single Bend

Problem 1

Given

$$Z_n = L_{2n} - 2n,$$

find

$$Z_n - Z_{n-1}.$$

Hence derive a recurrence relation for Z_n.

Problem 2

Given

$$Z_0 = 1,$$

and

$$Z_n = Z_{n-1} + 4n - 3,$$

find

$$Z_1, Z_2, Z_3, Z_4, Z_5.$$

Find the closed form of Z_n.

Problem 3

Compare

$$Z_n$$

and

$$L_{2n}.$$

Find

$$L_{2n} - Z_n.$$

Problem 4

Let

$$D_n = Z_n - L_n.$$

Find

$$D_1, D_2, D_3, D_4,$$

derive a recurrence relation for D_n, and find its closed form.

Find the smallest n such that

$$D_n \geq 100.$$

HW
Concrete Mathematics, p. 18, 8-11